# ARTIFICIAL INTELLIGENCE LABORATORY

## Comprehensive Technical Portfolio & Academic Defense

Course Code: **IC-709D** | Course Title: **Artificial Intelligence**

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**Academic Term:** Academic Year 2026–2027 | **Implementation Language:** Python 3.8+

# Student & Institutional Information

Academic Affiliation, Course Identification & Repository Invariants

## Institutional Profile

- **University:** Devi Ahilya Vishwavidyalaya (DAVV), Indore
- **Institute:** International Institute of Professional Studies (IIPS)
- **Department:** Department of Computer Science & Engineering
- **Program:** Integrated Master's / Postgraduate Engineering
- **Academic Session:** Academic Year 2026–2027
- **Location:** Indore, Madhya Pradesh, India

## Curricular Specifications

- **Course Code:** IC-709D
- **Course Title:** Artificial Intelligence
- **Primary Focus:** Problem Solving, Knowledge Representation, Machine Learning, Deep Learning, Explainable AI

## Student Identification

- **Candidate Name:** Jeet Verma
- **Roll Number:** IC-2K23-46
- **Repository:** IC-2K23-46-JEET-VERMA

## Development & Execution Environment

- **Language Runtime:** Python 3.8+ (Native Linux x86_64)
- **Numerical & Vision Libraries:** NumPy, Pandas, Scikit-Learn, PyTorch, Torchvision
- **Plotting & Diagnostics:** Matplotlib, Seaborn
- **Verification Standards:** Strictly zero data fabrication; empirical metrics derived directly from executed laboratory source code.

# Introduction to Artificial Intelligence

## Evolution of AI: From Classical Symbolic Reasoning to Differentiable Deep Learning

### Foundational Paradigms of AI

Artificial Intelligence (AI) encompasses computational principles that enable machines to perceive environments, formulate goals, reason symbolically, and optimize actions.

- **Symbolic AI (GOFAI):** Explores explicit declarative models using formal logic, semantic nets, and production rules with sound, verifiable inference chains.
- **State-Space Search:** Formulates discrete problem environments as state graphs, finding optimal operator sequences via BFS/DFS traversal.
- **Sub-symbolic / Statistical AI:** Learns non-linear representations and statistical associations directly from empirical data distributions.
- **Deep Differentiable Learning:** Employs automatic differentiation and gradient-based backpropagation through parameter manifolds.

### Modern AI Engineering Challenges

Deploying AI systems in safety-critical domains introduces essential engineering imperatives explored in this laboratory:

- **Data Leakage Prevention:** Strict isolation of training transformations from test evaluation folds.
- **Model Interpretability (XAI):** Mitigating 'black-box' opacity in deep CNNs via Grad-CAM class activation localization.
- **Adversarial Robustness:** Evaluating vulnerability to high-dimensional perturbations synthesized via Fast Gradient Sign Method (FGSM).
- **Subspace Compression:** Reducing dimensionality via mathematically equivalent PCA and SVD factorizations.

# Laboratory Objectives

## Core Curricular Goals & Competency Outcomes for Course IC-709D

| Objective Domain | Key Technical Competency | Implementation Framework |
|---|---|---|
| **1. Problem Solving** | Formulate state spaces, define valid transition operators, and execute graph search algorithms. | Python Standard Library (`collections.deque`, sets) |
| **2. Graph Search** | Implement BFS and DFS for grid pathfinding, puzzle solving, and road route planning. | Adjacency Lists, Queue & Recursion Stack |
| **3. Data Engineering** | Build leak-free pipelines: missing-value imputation, one-hot encoding, and ANOVA/MI selection. | NumPy, Pandas, Scikit-Learn Preprocessing |
| **4. Knowledge Representation** | Model domain entities using Predicates, Semantic Networks, Frames, and Forward Chaining. | Declarative KB, Inheritance DAGs, Rule Engine |
| **5. Supervised Learning** | Train white-box Decision Trees, mitigate overfitting via depth pre-pruning, inspect Gini importance. | Scikit-Learn `DecisionTreeClassifier` |
| **6. Deep Learning** | Construct PyTorch computational DAGs, execute reverse-mode Autograd, optimize MSE/BCE losses. | PyTorch (`torch.nn`, `torch.optim`, `autograd`) |
| **7. Advanced AI Topics** | Implement Grad-CAM XAI hooks, synthesize FGSM attacks, prove algebraic PCA $\equiv$ SVD. | PyTorch Hooks, CIFAR-10 Vision, NumPy SVD |

# Overview of All Seven Laboratory Units

## End-to-End Progression Across AI Methodologies

| | | |
|---|---|---|
| **Lab 1: AI Problem Solving**<br>Water Jug (4L/3L), Vacuum Cleaner Agent, Missionaries & Cannibals (3M/3C), 8-Puzzle State Space Representation. | **Lab 2: Search & Navigation**<br>8-Puzzle Solver (BFS), Graph Traversal (BFS & DFS), Grid Maze Solver (DFS), Robot Navigation (BFS), Route Planning (BFS). | **Lab 3: AI Frameworks & Data Prep**<br>Leak-free tabular pipeline on Campus Placement & Bank datasets; median imputation, one-hot encoding, ANOVA & Mutual Info. |
| **Lab 4: Knowledge Representation**<br>Predicate Logic inference (AND/OR/NOT), Semantic Network inheritance, Minsky Frame hierarchies, Forward-Chaining Rule Engine. | **Lab 5: Decision Tree Classifier**<br>CART induction on Placement data; target leakage elimination, Gini impurity splits, $69.77\%$ test acc, depth-2 pre-pruning ($79.07\%$). | **Lab 6: Deep Learning in PyTorch**<br>Tensors, Autograd DAGs, Parametric Linear Regression ($R^2=0.9619$), MLP non-linear classifier on Moons data ($95.0\%$ acc). |
| **Lab 7: Explainable AI, Adversarial AI & Advanced Applications**<br>• **Grad-CAM:** Dynamic PyTorch hooks + GAP heatmap attribution on CIFAR-10.<br>• **FGSM Attack:** $\ell_\infty$ adversarial perturbation; accuracy collapse from $38\%$ to $4\%$ at $\epsilon=0.05$.<br>• **PCA & SVD:** Unsupervised dimensionality reduction on Iris ($95.81\%$ variance); algebraic equivalence proof ($\Delta < 10^{-14}$). | | |

# Intelligent Agents and Problem-Solving (Lab 1)

## Agent-Environment Architecture, State Representation & Constraint Satisfaction

### Vacuum Cleaner Agent Model

- **Environment:** Discrete two-room world (Room A, Room B).
- **Percept Vector:** `(Location, Status_A, Status_B)` where Location $\in \{'A', 'B'\}$, Status $\in \{0, 1\}$.
- **Actuators:** `SUCK`, `MOVE_LEFT`, `MOVE_RIGHT`.
- **Search Strategy:** BFS systematically cleans both rooms in minimal 3 steps from initial state `('A', 0, 0)`.

### Water Jug & River Crossing Constraints

- **Water Jug Problem:** State `(a, b)` with capacities $(4, 3)$, goal $2$L in either jug. Solved via BFS in 4 valid pours without measuring scale.
- **Missionaries & Cannibals:** State `(M, C, Boat)` with safety invariant $M \ge C$ or $M=0$. BFS prunes unsafe states and finds the unique 11-step safe crossing.

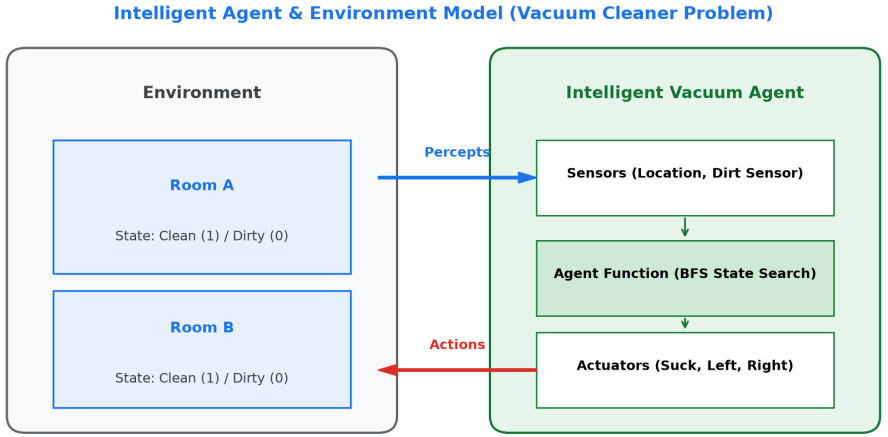



Figure: Intelligent Agent & Environment interaction model (Vacuum Cleaner).

# State-Space Representation (Lab 1 & Lab 2)

## Mathematical Formalization of AI State Spaces & 8-Puzzle Successors

### Formal Definition of State Space

A classical AI problem space is formally defined as a 4-tuple $\mathcal{P} = \langle S, s_0, \mathcal{O}, G \rangle$:

1. $S$: The set of all possible configurations (state space).
2. $s_0 \in S$: Initial state configuration.
3. $\mathcal{O}: S \to \mathcal{P}(S)$: Allowable transition operators.
4. $G \subseteq S$: Goal predicate or target configuration set.

### 8-Puzzle Formulation

- **Board:** $3 \times 3$ grid with tiles $1..8$ and blank space $0$.
- **State Space Size:** $|S| = \frac{9!}{2} = 181,440$ reachable states.
- **Operators:** Blank tile moves $\{\text{Up}, \text{Down}, \text{Left}, \text{Right}\}$ subject to grid boundaries.
- **Verification:** From `[1 2 3 / 4 0 6 / 7 5 8]`, BFS generates 4 valid successors and reaches goal `[1 2 3 / 4 5 6 / 7 8 0]` in 2 moves.

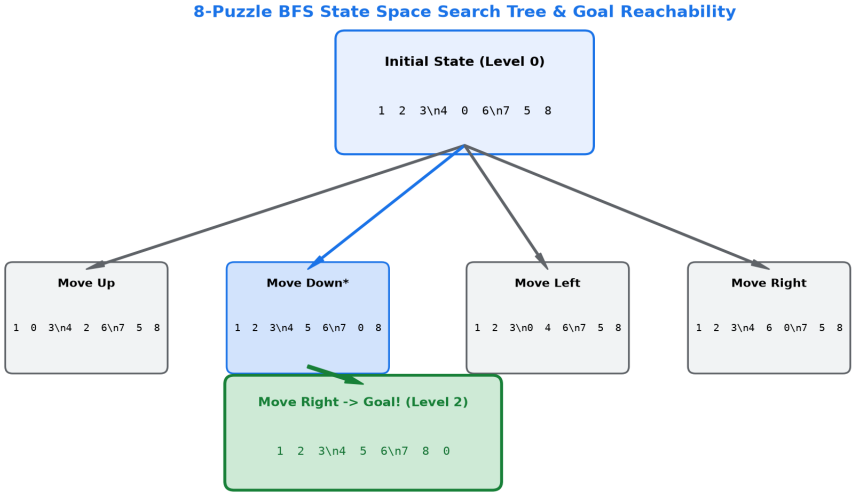



Figure: 8-Puzzle BFS State Space Search Tree and 2-step solution.

# Uninformed Search Algorithms (Lab 2)

## Graph Traversal, Grid Maze Backtracking & Shortest Route Planning

### Breadth-First Search (BFS)

- **Mechanism:** Level-order FIFO queue exploration.
- **Optimality:** Guarantees shortest path in unweighted graphs.
- **Complexity:** Time $O(V + E) = O(b^d)$, Space $O(b^d)$.
- **Applications:** Route planning ($A \to B \to D \to G$ in 3 hops), Robot grid navigation, 8-Puzzle solving.

### Depth-First Search (DFS)

- **Mechanism:** Recursive LIFO exploration along branch before backtracking.
- **Memory Advantage:** Linear space $O(b \cdot m)$ requires vastly less RAM than BFS.
- **Trade-off:** Non-optimal; may explore deep suboptimal subtrees.
- **Applications:** Grid Maze Solver obstacle navigation, cycle detection.

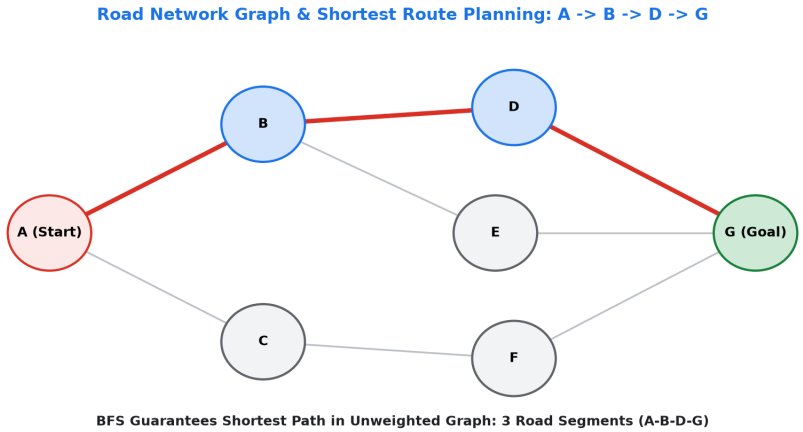



Figure: BFS Route Planning over road network graph (A -> B -> D -> G).

# Informed Search Algorithms: Theoretical Trade-offs

## Heuristic Search ($A^*$, Greedy Best-First) vs. Implemented Uninformed Search

### The Curse of Exponential Frontier Growth

In large state spaces (e.g. 15-Puzzle $|S| \approx 10^{13}$), blind uninformed search (BFS) fails due to exponential memory exhaustion ($O(b^d)$).

- **Heuristic Function $h(n)$:** Estimates cost from state $n$ to closest goal.
- **Admissibility:** $h(n) \le h^*(n)$ (never overestimates true cost).
- **Consistency (Monotonicity):** $h(n) \le c(n, a, n') + h(n')$.

### Heuristic Search Strategies

- **Greedy Best-First Search:** Expands node with minimal $h(n)$. Highly efficient but incomplete in infinite loops and non-optimal.
- **$A^*$ Search:** Minimizes $f(n) = g(n) + h(n)$. Unconditionally optimal and optimally efficient under consistent heuristics.

### Curricular Implementation Invariant

**Repository Scope Clarification:**
The laboratory problem-solving units (Lab 1 & Lab 2) specifically focused on foundational **Uninformed Search (BFS & DFS)** because the state graphs featured unit-cost transitions ($c=1$), where BFS strictly guarantees optimal minimal-hop solutions without requiring heuristic domain estimates.

### Complexity Comparison Matrix

| Strategy | Time | Space | Optimal |
|---|---|---|---|
| **BFS** | O(b^d) | O(b^d) | Yes (c=1) |
| **DFS** | O(b^m) | O(b·m) | No |
| **Greedy** | O(b^m) | O(b^m) | No |
| **A*** | O(b^{d*}) | O(b^{d*}) | Yes (Admissible) |

# Knowledge Representation and Reasoning (Lab 4)

## Symbolic AI: Predicates, Semantic Graphs, Frames & Forward Chaining

### Symbolic Knowledge Representation Paradigms

- **Predicate Logic:** First-order assertions $P(t_1..t_n)$, exact & wildcard unification queries (`ask`/`query`), and truth connectives (AND, OR, NOT) for academic eligibility auditing.
- **Semantic Networks:** Associative graphs with `IS-A`, `HAS-A`, `PART-OF` links. Property inheritance traverses upward along DAG taxonomies.
- **Minsky Frame Systems:** Object-centered schemas with slots, facet defaults, and instance attribute overrides.
- **Forward-Chaining Rule Engine:** Production system matching active facts against rule antecedents, deducing new assertions across cycles until fixed-point saturation.

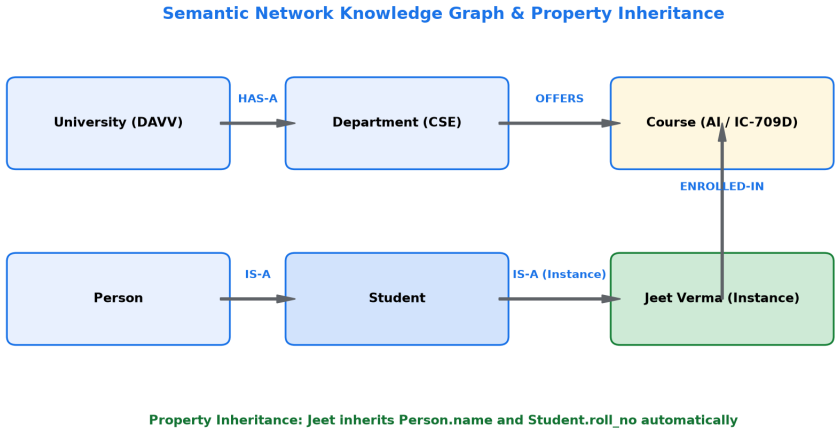



Figure: Semantic Network knowledge graph and property inheritance.

# Machine Learning & Implemented AI Techniques (Labs 3, 5, 6, 7)

Tabular Preprocessing, CART Decision Trees, PyTorch Autograd & Dimensionality Reduction

### Data Prep & Decision Trees (Labs 3 & 5)

- **Leak-Free Pipeline:** Stratified 80:20 split, numerical median imputation, categorical one-hot encoding fitted strictly on training data.
- **Statistical Feature Selection:** ANOVA $F$-test (`f_classif`) and Mutual Information (`mutual_info_classif`).
- **CART Decision Tree:** Binary recursive partitioning minimizing Gini Impurity. Achieved $69.77\%$ test accuracy unpruned, and $79.07\%$ with `max_depth=2` pre-pruning.
- **Dominant Predictor:** Secondary education score (`ssc_p`) accounted for $44.78\%$ of total Gini importance.

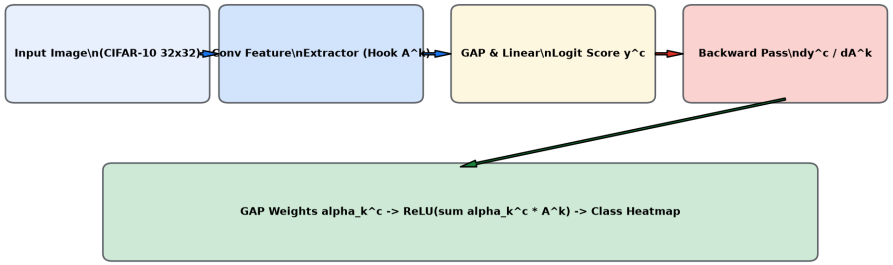
### Deep Learning & Advanced AI (Labs 6 & 7)

- **PyTorch Dynamic Autograd:** Reverse-mode automatic differentiation via directed acyclic graphs (DAGs).
- **Linear Regression:** Converged to MSE $1.542$ and $R^2 = 0.9619$ over 150 epochs under SGD.
- **Non-Linear MLP:** Moons classification via `Linear(2, 16) -> ReLU -> Linear(16, 1)`, reaching $95.0\%$ accuracy.
- **PCA & SVD Equivalence:** Iris dimensionality reduction; PC1+PC2 capture $95.81\%$ variance; verified algebraic identity $\lambda_i = \frac{s_i^2}{N-1}$ with $\Delta < 10^{-14}$.

# Important Diagrams & Flowcharts (Lab 7)

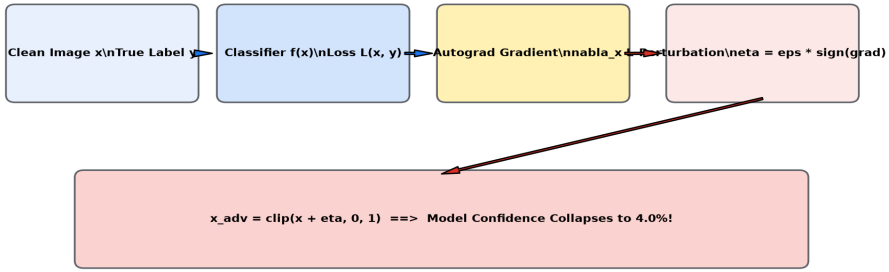
## Explainable AI (Grad-CAM) & Fast Gradient Sign Method (FGSM) Pipelines

**Grad-CAM: Gradient-Weighted Class Activation Mapping Workflow (Lab 7)**




**Grad-CAM Pipeline:** Forward CNN pass, backward hook gradient capture $\frac{\partial y^c}{\partial A^k}$, Global Average Pooling weights $\alpha_k^c$, and ReLU class activation heatmap overlay.

**Fast Gradient Sign Method (FGSM) Adversarial Generation Pipeline**




**FGSM Adversarial Pipeline:** Autograd input gradient $\nabla_x \mathcal{L}$, sign extraction, $\ell_\infty$ scaling by $\epsilon$, and clipping to synthesize high-confidence adversarial misclassifications.

# Complexity Analysis Summary

## Comparative Space-Time Bounds Across Implemented Algorithmic Paradigms

| Paradigm | Algorithm / Experiment | Time Complexity | Space Complexity | Theoretical Guarantee |
|---|---|---|---|---|
| **Uninformed Search** | Water Jug / 8-Puzzle (BFS) | O(b^d) | O(b^d) | Optimal shortest path (c=1) |
| **Uninformed Search** | Maze Solver (DFS) | O(b^m) | O(b · m) | Linear memory advantage |
| **Symbolic KR** | Semantic Network / Frames | O(V + E) / O(H · S) | O(V + E) | Sound inheritance traversal |
| **Rule Inference** | Forward Chaining Engine | O(R · \|F\|^k) | O(\|F\|) | Fixed-point saturation |
| **Decision Trees** | CART Classification (Lab 5) | O(p · N log N · d) | O(2^d) tree nodes | Axis-parallel partitioning |
| **Deep Learning** | PyTorch Autograd & SGD | O(E · N · W) | O(W) + O(N) DAG | Reverse-mode gradient chain rule |
| **Explainable AI** | Grad-CAM XAI (Lab 7) | O(FLOPs_fwd + FLOPs_bwd) | O(C · H · W) | Single backward pass localization |
| **Adversarial AI** | FGSM Perturbation (Lab 7) | O(FLOPs_bwd) = O(1) pass | O(C · H · W) | Max loss increase along sign(grad) |
| **Dimensionality** | PCA & SVD Reduction (Lab 7) | O(min(N^2 D, N D^2)) | O(N · D) | Optimal low-rank variance |

# Experimental Observations and Quantitative Results

## Empirical Findings Across Verified Laboratory Executions

### Verified Experimental Observations

- **Search Optimality (Labs 1 & 2):** BFS strictly identified the minimal 4-pour Water Jug solution, 11-step safe river crossing, and 2-move 8-Puzzle solution (`Down`, `Right`).
- **Target Leakage Impact (Lab 5):** Dropping post-outcome variable `salary` prevented catastrophic target leakage, yielding true test accuracy of $69.77\%$ ($75.76\%$ precision, $83.33\%$ recall).
- **Decision Tree Pruning (Lab 5):** Constraining tree depth to `max_depth=2` resolved variance inflation, boosting test accuracy from $69.77\%$ to $79.07\%$.
- **Deep Regression Fit (Lab 6):** PyTorch SGD minimized MSE loss from $145.54$ down to $1.39$ ($R^2 = 0.9619$), matching the synthetic noise floor $\sigma^2 = 1.44$.
- **Adversarial Vulnerability (Lab 7):** FGSM attack on CIFAR-10 at $\epsilon=0.05$ caused accuracy to collapse from $38.0\%$ to $4.0\%$.
- **Subspace Equivalence (Lab 7):** PCA and SVD projections yielded identical coordinates ($\|X_{\text{PCA}} - X_{\text{SVD}}\|_\infty = 4.0 \times 10^{-15}$).

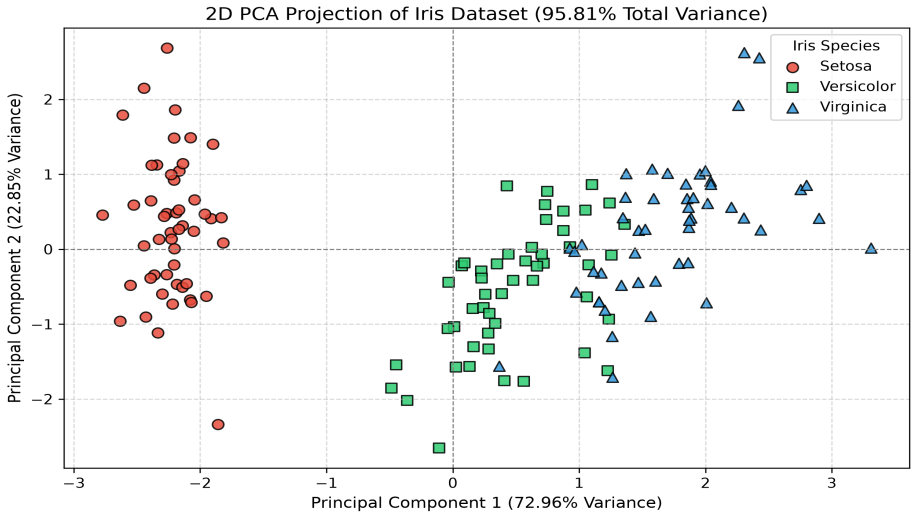



Figure: Actual 2D PCA projection of Iris dataset ($95.81\%$ variance).

# Practical Applications in Industry & Research

## Translating Laboratory Implementations into Real-World Deployments

| | |
|---|---|
| **Autonomous Navigation & Robotics**<br>Grid-based BFS/DFS algorithms form the core of AGV warehouse routing, drone pathfinding, and vacuum robot coverage maps. | **Automated Logical Auditing**<br>Predicate logic and production rule engines power university degree clearance, legal compliance checkers, and medical triage decision support. |
| **Institutional Recruitment Analytics**<br>Interpretable CART decision trees enable university career centers to identify at-risk students early and design targeted skill-building interventions. | **Differentiable Neural Control**<br>PyTorch tensor computation and dynamic autograd underpin production deep learning systems in time-series forecasting, vision, and natural language. |
| **Explainable Medical Imaging (XAI)**<br>Grad-CAM visual localization ensures clinical accountability by verifying that diagnostic models attend to genuine pathological lesions rather than imaging artifacts. | **AI Security & Adversarial Defense**<br>FGSM vulnerability auditing guides adversarial training, robust feature regularization, and input sanitation in biometric and financial verification systems. |

# Learning Outcomes & Skill Mastery

## Key Technical Competencies Acquired Throughout Course IC-709D

### Theoretical & Algorithmic Foundations

- Formulate real-world problems mathematically into state-space search graphs.
- Analyze asymptotic space-time bounds and evaluate completeness/optimality guarantees.
- Implement declarative predicate knowledge bases and understand closed-world reasoning.
- Understand forward chaining production cycles and fixed-point termination.
- Grasp the mathematical duality connecting Covariance PCA and SVD factorizations.

### Practical Engineering Competencies

- Construct leak-free machine learning pipelines using Scikit-Learn transformers.
- Eliminate target leakage and evaluate models using comprehensive multi-class metrics.
- Engineer modular neural networks in PyTorch, executing canonical 5-step training loops.
- Intercept convolutional feature maps and gradients using PyTorch hooks for Grad-CAM.
- Synthesize adversarial examples via Autograd input-gradient ascent to audit model resilience.

# Conclusion

## Summary of Laboratory Work & Methodological Synthesis

**Comprehensive AI Coursework Synthesis:**
The **IC-709D: Artificial Intelligence** laboratory successfully bridges classical symbolic artificial intelligence, discrete state-space search, data engineering pipelines, interpretable machine learning, and modern differentiable deep learning.

**Key Takeaways:**
1. **Algorithmic Rigor:** BFS guarantees shortest-path optimality in unweighted spaces; DFS conserves memory at the cost of path length.
2. **Engineering Guardrails:** Target leakage and unstandardized scales completely invalidate machine learning models if not strictly mitigated via training-only pipeline fitting.
3. **Interpretability & Pruning:** Tree depth pre-pruning ($d=2$) boosts placement test accuracy to $79.07\%$; Grad-CAM provides critical visual rationale for deep convolutional predictions.
4. **Security & Geometry:** Deep models are fragile to gradient-aligned adversarial perturbations (FGSM collapses accuracy to $4.0\%$); PCA and SVD provide exact algebraic equivalence for dimensionality reduction.

# References & Academic Literature

## Authoritative Textbooks and Foundational Research Papers

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